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Seasonal dynamics of physico-chemical characteristics and biological responses of Lake Chilwa, Southern Africa

Messias A. Macuiane [a,b,*], Emmanuel K.W. Kaunda [a,1], Daniel Jamu [b,2]

[a] University of Malawi/Bunda College of Agriculture, Department of Aquaculture and Fisheries Sciences, PO Box 219, Lilongwe, Malawi
[b] WorldFish Center, PO Box 229, Zomba, Malawi

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ABSTRACT

Lake Chilwa is shared by Malawi and Mozambique, it supports an important fishery and its watershed is undergoing rapid population growth and increasing utilization for agricultural production. It is a shallow, closed basin lake with extensive surrounding wetlands; and it has suffered several desiccation events in the last century. To better understand the current condition of the lake, we monitored a suite of physical, chemical and biological parameters at approximately monthly intervals over an annual cycle in 2004–2005. The limnology of the lake was extremely sensitive to seasonal changes in the lake's seasonal hydrological cycle. The physico-chemical parameters, temperature, electrical conductivity, and total suspended solids exhibited clear seasonal patterns driven by the highly seasonal rainfall and resultant lake levels. In response, phytoplankton and zooplankton abundance, as well as biologically dependent oxygen concentrations and pH, exhibited several maxima levels over the year. The peaks of phytoplankton and zooplankton were out of phase suggesting a lag in the zooplankton grazing in response to pulses in primary productivity. Chlorophyll concentrations can exceed 1 mg/L in surface waters indicative of hypereutrophic conditions, but they fell dramatically during zooplankton peaks. This hydrologically driven, shallow and mesohaline lake is a productive and critical resource to the region. Its management poses challenges arising from the dependence of its limnology and fishery on the lakes hydrology, catchment land use and climate variability.

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Introduction

Lake Chilwa is a closed basin (endorheic), transboundary lake receiving inflows from Malawi and Mozambique (Fig. 1). The close linkage between catchment and lake which is a fundamental paradigm in limnology is even stronger in endorheic lakes where all nutrients and other materials washed out from the catchment are retained in the lake and its sediments (Aladin and Potts, 1992; Coe and Foley, 2001). Population growth in Malawi is causing increasing pressure on the land catchment from intensive agricultural practices, including increasing use of inorganic fertilizers, that may have detrimental influence on Lake Chilwa by altering its already sensitive water budget and increasing its sediment and nutrient budgets to the possible detriment of the it valuable fishery.

Lake Chilwa has long been recognized as being among the most productive in Africa on a real basis (e.g. Bayley, 1988). The importance of Lake Chilwa fishery for both Malawi and Mozambique was recognized for the first time in 1963 when it supplied an annual catch of 9000 tons just in Malawi (Kalk et al., 1979) where it provides on average over 20% of the total catch for the country and an estimated annual income of about 800 million Malawi Kwacha ($10 million) to its fishing communities (Environmental Affairs Department, 2001). Although the lake is shared with Mozambique, similar statistics there are not available because of the lake's remoteness from major population centers there. In some years, the annual catch can approach 25,000 tons with a long term average of around 13,000 tons landed, primarily in Malawi (Environmental Affairs Department, 2000). In 1979 fish production from Lake Chilwa contributed approximately 43% of national landings, while in 1990 its contribution was 33% (Environmental Affairs Department, 2000).

Early studies of Lake Chilwa (e.g. Kalk et al., 1979) emphasized the importance of the surrounding large area of swamp and floodplain in contributing to the high fish productivity of the lake (Howard-Williams and Lenton, 1975; Cantrell, 1988; Environmental Affairs Department, 2000). Half of the lake area is occupied by permanent *Typha domingensis* vegetation (Howard-Williams and Walker, 1974) that provides detritus to the lake (Howard-Williams and Walker, 1974; Kalk, 1979). The vegetation was believed to be a primary source of trophic energy and nutrients that support a productive food web in Lake Chilwa (Howard-Williams and Lenton, 1975; Kalk 1979). The surrounding wetlands also provide

* Corresponding author. University of Malawi/Bunda College of Agriculture, Department of Aquaculture and Fisheries Sciences, PO Box 219, Lilongwe, Malawi. Tel.: +265 999285590.
E-mail addresses: mmacuiane@yahoo.co.uk (M.A. Macuiane), ekaunda@yahoo.com (E.K.W. Kaunda), d.jamu@cgiar.org (D. Jamu).
[1] Tel.: +265 999510796.
[2] Tel.: +265 888821701.

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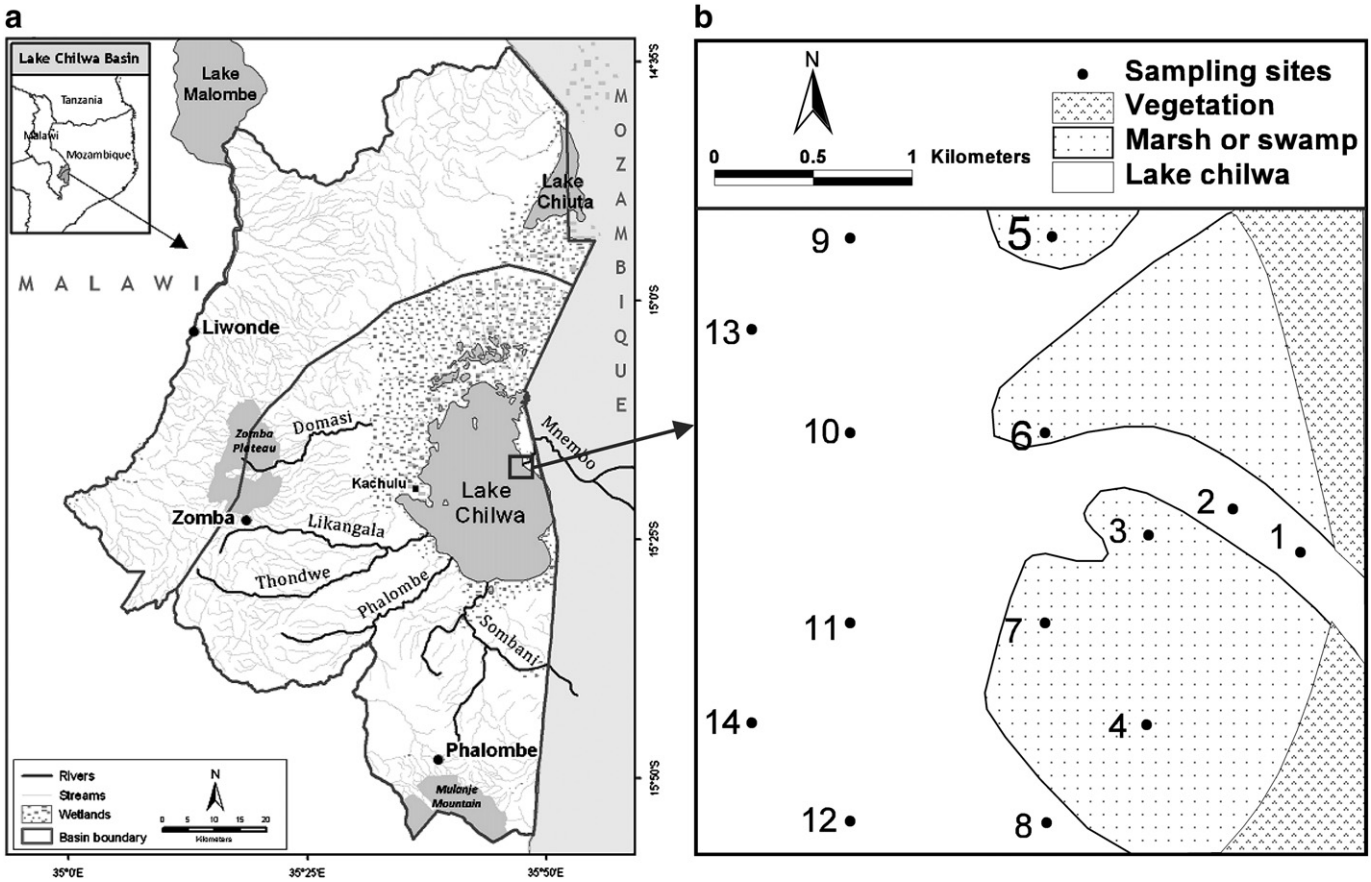

Fig. 1. Map of (a) Lake Chilwa basin showing Major towns and rivers (in italics) and (b) the study sampling sites. Sites 1 and 2 are in the Mnembo River channel.

breeding grounds for highly valued fish species such as *Clarias gariepinus* and *Barbus* spp. (Kalk, 1979; Macuiane et al., 2009) as well as nursery and feeding grounds for juveniles of these and other species.

The primary species contributing to the fishery, *Barbus paludinosus* Peters, *Oreochromis shiranus chilwae* Trewavas and *Clarias gariepinus* Peters with *Barbus* contributing most of the landings by weight while *Clarias* and *Oreochromis* fetch higher prices. *B. paludinosus* is a zooplanktivore throughout its life history and feeds primarily on zooplankton in the lake (Kalk, 1979). The importance of zooplankton to the three main species of the fishery was demonstrated by Bourn (1973) who investigated the gut content of the juveniles of the three species and found that they are largely planktivorous having respectively 60%, 69%, and 48% zooplankton in their guts. The zooplankton are phytoplanktivores and facultative detritus feeders (Kalk, 1979), and so their abundance is dependent on a rich food supply which is maintained by authtochthonous primary productivity as well as detrital contributions of the surrounding *Typha* wetlands (Kalk, 1979).

Fisheries production and landings in Lake Chilwa, although high on average, is not constant and is directly affected by fishing pressure and water levels (Furse et al., 1979; Fig. 2). Low water levels do recur in response to drought years with catastrophic consequences for fish production. If the rainfall is below average for a succession of two to three years, the lake may dry completely with water only remaining in the river mouths and in some pools in the wetland. Annual water fluctuations can vary as much as 2–3 m, but can be amplified in dry

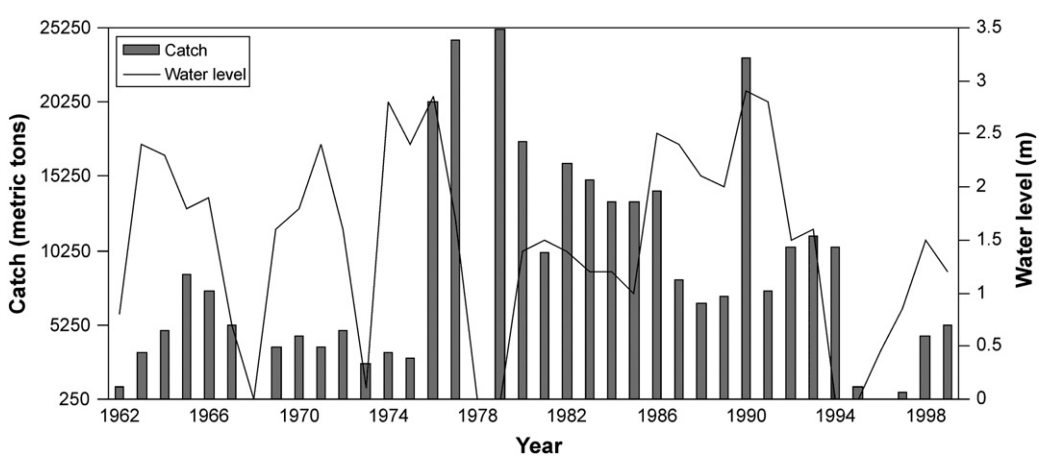

Fig. 2. Annual catches (shaded bars) and lake levels in Lake Chilwa, Malawi 1962-1999. Source: The water level provided by Environmental Affairs Department (2000), and Fisheries data from Department of Fisheries, GoM (1999). Continuous water level records are unavailable after 1999 due to loss of recording equipment.

years and lead to rapid and complete desiccation of the lake (Lancaster, 1979). The lake went through moderate level recessions during the years 1884, 1900/20/23, 1931/33/34, 1943/49, 1953/55, and 1960/61 with severe recession resulting in a dry lake bed in 1914/15 and 1967 (Morgan, 1971) and 1995 (Ambali and Kabwazi, 1999). There has not been a severe water level recession since 1995.

Over the past century minor recessions have occurred on average every six years. However, the basin is able to refill rapidly after heavy rains and fish stocks in the lake can recover within three to four years (Njaya, 2001). Recolonization relies on founding populations which survive in the rivers and pools in the wetland during periods of severe and moderate recessions respectively (Ambali and Kabwazi, 1999). Over recent decades, the number of fishermen and fishing gears has increased tremendously due to human population growth. For instance, the lake supported 2,000 fishermen before 1967 (Morgan, 1971) and by 1995 it supported over 36,000 fishermen and middle-men (Kabwazi and Wilson, 1998) and the number likely continues to increase as the fishery is open access. The increasing number of fishers magnifies the social and economic consequences of recurring drought which may increase with global climate change in this part of Africa (Hulme et al., 2001). The long term combination of increasing fishing pressure and generally declining water levels has resulted in the decline of annual per capita fish supply to 4.2 kg today for the people in the vicinity of the lake which is far below 15 kg per capita fish consumption recommended by the World Health Organization.

There were several previous limnological studies from 1960s and 1970s (Moss and Moss, 1969; McLachlan and McLachlan, 1969; Morgan and Kalk, 1970; McLachlan et al., 1972; Howard-Williams, 1972; Lancaster, 1979); but none characterized the physico-chemical parameters of the lake in the vicinity of the inflowing rivers which are considered critical nutrient sources as well as breeding areas for commercial species. Jamu and Brummett (1999) investigated the migration and reproduction of *Barbus* species in relation to physico-chemical parameters in Domasi and Likangala rivers, flowing from Malawi catchment into Lake Chilwa; however, no studies have been conducted on the Mozambican waters prior to this study.

The paucity of information on the rivers and their influence on the lake is regrettable as the river-lake continuum is important to the ecology of fish species that make up the bulk of fish catches. Moreover, the increasing impact of human activities may change the physico-chemical condition of the rivers rapidly, and the impacts may differ among catchments. Malawi has high population density while the Mozambique catchment is more sparsely populated, but there are relatively steep slopes in the Chilwa catchment headwaters in both Malawi and Mozambique which can contribute to high sediment run-off, particularly as the slopes are brought into cultivation. Therefore, in order to formulate a holistic management strategy for Lake Chilwa catchment the present study investigates the annual variation of physico-chemical and biological characteristics of L. Chilwa and a major inflow, the Mnembo River, from four different sampling sites: the Mnembo River mouth, intermediate deltaic areas between the river mouth and open lake, fringing areas of rooted emergent vegetation and the open lake to provide an incremental step towards a better understanding of the lake function and improve its management.

Materials and methods

Study area

Lake Chilwa is a closed saline lake in southeastern Africa, shared by Malawi and Mozambique (Boscce and Bernacscek, 1990) that drains a catchment that is increasingly important as agricultural land. In general, Lake Chilwa is shallow ($\leq$6 m) with an open water area of up to 680 km^2 surrounded by 600 km^2 of *Typha* swamps, 390 km^2 of marshes and 580 km^2 of seasonally inundated floodplain (van Zwieten and Njaya, 2003). In Malawi, the lake is situated in southern region, about 50 km south of the rift valley and Lake Malawi (Niassa) (15° 30′S and 35° 35°45′E) while in Mozambique the lake is situated in the eastern side of its central region in Niassa Province (Fig. 1). There are five major influent rivers draining into Lake Chilwa, the Domasi, Likangala, Mnembo, Phalombe and Sombani, with maximum discharges during the rainy season of December to April, usually peaking in February (Lancaster, 1979). Jamu et al. (2003) suspect that Mnembo River, the only tributary for which no flow data are available and which flows from Mozambique, is actually the lake's largest source of water.

Field sampling

This investigation was undertaken from February 2004 to January 2005 during the day from 8 am to 4 pm, but sampling times for individual stations varied depending on lake conditions. Meteorological data (air temperature, rainfall and wind speed) were provided by the Malawi Department of Meteorology for a station at Ntanja, close to the lake and river mouth.

Limnological observations were made at an array of sampling stations from the Mnembo River to the open waters of Lake Chilwa (Fig. 1). The lake level change was recorded by measuring water depth from one site located on each sampling date by GPS, and these measurements were used as relative lake levels (Fig. 3c). The water depth recorded from this site was 2.9 m and the minimum was 2.0 m giving an annual fluctuation of 0.9 m for our year of observation. The maximum water level recorded from all sampled sites was 5 m (station 14 in December 2004) and the minimum was 1.1 m (station 4 in October 2004). Because the lake is relatively large and shallow with continuous mixing by wind and the inflowing river, our near-surface samples taken between 0.1 and 0.3 m were considered representative of the water column. The stations were chosen to cover the Mnembo River mouth, adjacent rooted emergent vegetation, the inshore areas between the mouth and the open lake and the open water of Lake Chilwa (Fig. 1). Two stations were chosen in the Mnembo River mouth (1 and 2), three stations (3, 4 and 5) representative of the areas of emergent vegetation, three stations were chosen in intermediate area between the river mouth and open lake (6, 7 and 8) and six in the open lake (9, 10, 11, 12, 13 and 14). The sampling stations were established using GPS and covered a total area of approximately 5.5 square km.

At each site, at approximately monthly intervals, dissolved oxygen concentration (DO), electrical conductivity (EC), water temperature (WT), and pH values were recorded using a multi-parameter water meter (WTW 340i kit). The probe was held 0.2 to 0.3 m below water surface. The Total suspended solids concentrations (TSS) were determined by collecting one water sample in 500 ml plastic bottles at three different water levels (0.1, 0.2, and 0.3 m) from each site to overcome the possibility of fine diurnal stratification giving misleading results from a single surface sample especially during algal blooms in this hypereutrophic lake. TSS was analyzed according to Standard Methods (American Public Health Association, 1989) and the average TSS calculated. Similarly three water samples from 0.1, 0.2, and 0.3 m from each site were also used for the estimation of phytoplankton abundance measured as chlorophyll-*a* concentration. Chlorophyll-*a* subsamples were kept in the dark during transportation to the laboratory and analyzed within 12 hours after collection. Phytoplankton were collected onto a Whatman GF/F glass-fiber filter, and chlorophyll-*a* was extracted using 90% acetone. Chlorophyll a concentration was determined from the optical density (absorbance) of extracts with a spectrophotometer following the procedures of Boyd (1979). Similar to the TSS, chlorophyll-*a* concentrations for each site were expressed as an average of the three samples collected from the three depths. Chlorophyll-*a* analysis was not undertaken in the first two months of sampling and began in April 2004.

Zooplankton samples were collected from approximately 0.1–0.2 m with a 20 L bucket tied with a rope, and then concentrated by passing through an 80 μm mesh net to concentrate the larger zooplankton taxa. The material retained in the net was put into the 500 mL plastic bottle and 10 mL of 70% ethanol (American Public Health Association, 1989) was added into the sample to preserve the zooplankton. One mL of the sample was pipetted and placed in a Sedgwick-Rafter cell (Ludwig and Tackett, 1991) and counted for the calanoid copepod *Tropodiaptomous kraepelini* and the cladocerans, *Daphnia barbata*, *Diaphanasoma excism*, *Moina micrura,* and *Cerodaphnia cornuta.* For analysis of seasonal changes in water characteristics, the individual sites were grouped by zone: Mnembo River mouth (2 sites), area of emergent vegetation (3 sites), intermediate area between river mouth and open lake (3 sites) and open lake (6 sites) and averages and standard errors were calculated. Statistical differences among sites were not performed due to the low variability with the different zone and often high similarity among the zones on most sampling dates for the parameters measured.

Results

Air and water temperatures

Air and water temperature had very similar seasonal trends (Figs. 3b and 4b). The air temperature (monthly mean of daily mean of min and max) was high in February and declined to minimum values in July before increasing to another maximum in January. The air temperature values were lower than those of water temperatures as a result of the higher heat capacity of water. Low water temperatures (21.7 to 23.2 °C) were recorded between April and September and higher values (26.7 to 28.7 °C) were recorded between November and March. Water temperature lagged behind air temperature, e.g. reaching a maximum in January rather than October. Overall there was little difference in water temperature among the sampling sites, all zones showing the same seasonal pattern. However, a slight difference was observed in February and March, towards the end of the rainy season where the river mouth had slightly lower water temperatures compared to other areas (Fig. 4b). These slight differences may just be a consequence of the time of day of sampling as diurnal variation in temperatures would be expected in this shallow tropical lake.

Rainfall, relative water level and electrical conductivity

We started our sampling in February towards the end of the rainy season, and the rain stopped in May after three samplings. The no rain fell between May and September, the extended "dry season" period characteristic of Malawi, and the next rainfall initiated the 'rainy season' in October. Substantial amounts of rainfall were recorded through February (112.3 mm) and March (98.7 mm), but highest rainfall occurred in December (217.6 mm), and January (227.3 mm) (Fig. 3a).

The rainy season begun in October when the lake was at its lowest relative water level having been declining through the dry season due to evaporative losses (2 m at our recording site) after which the level (Fig. 3c) rises showing the dependence of lake level on rainfall. With the cessation of the rains, relative water levels again fell from a high of 2.8 m at the recording site in March to lowest relative levels at the end of dry season.

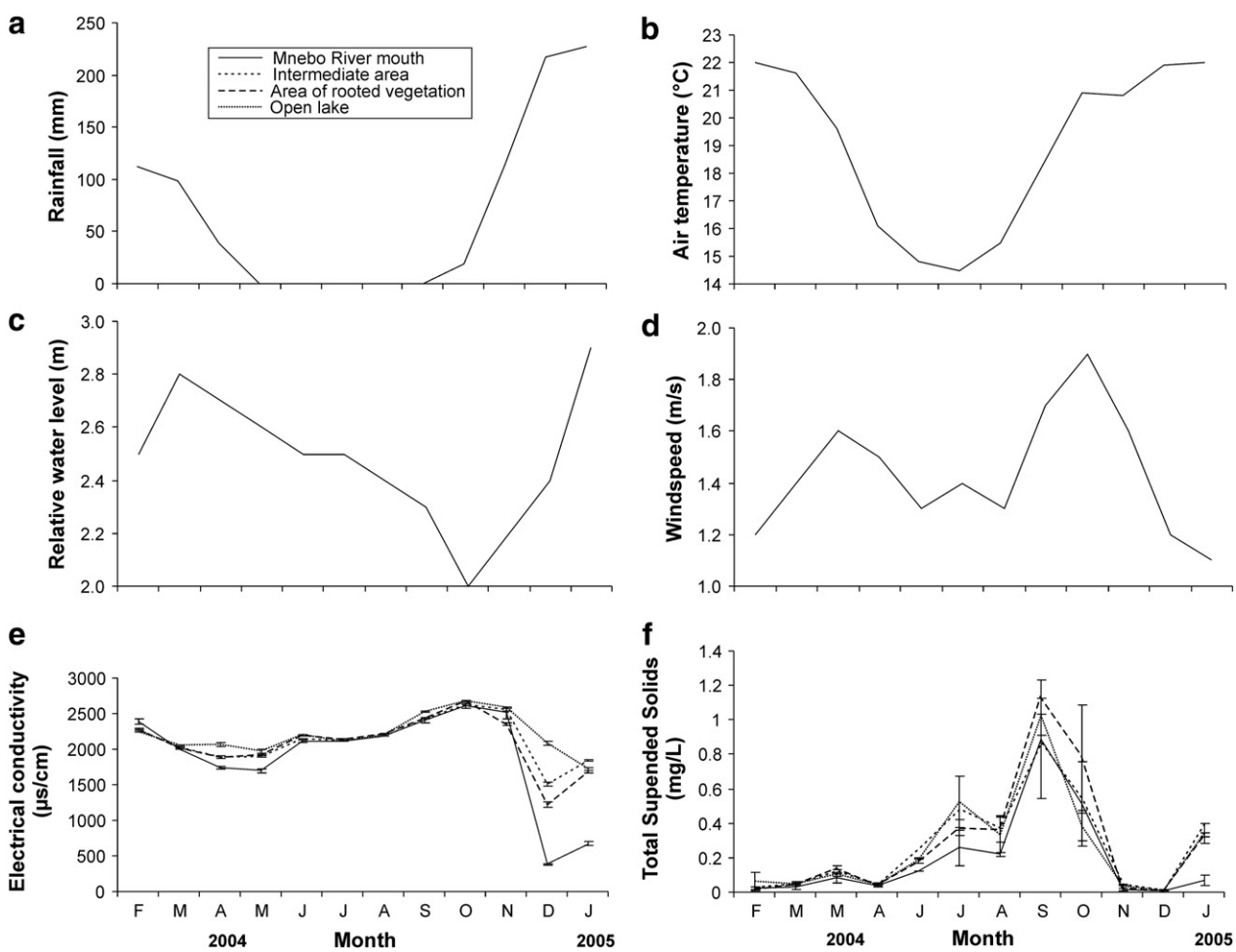

Fig. 3. Seasonal variation of (a) rainfall, (b) air temperature, (c) relative water level, (d), windspeed, (e) electrical conductivity, and (f) total suspended solids. Mean ± SE.

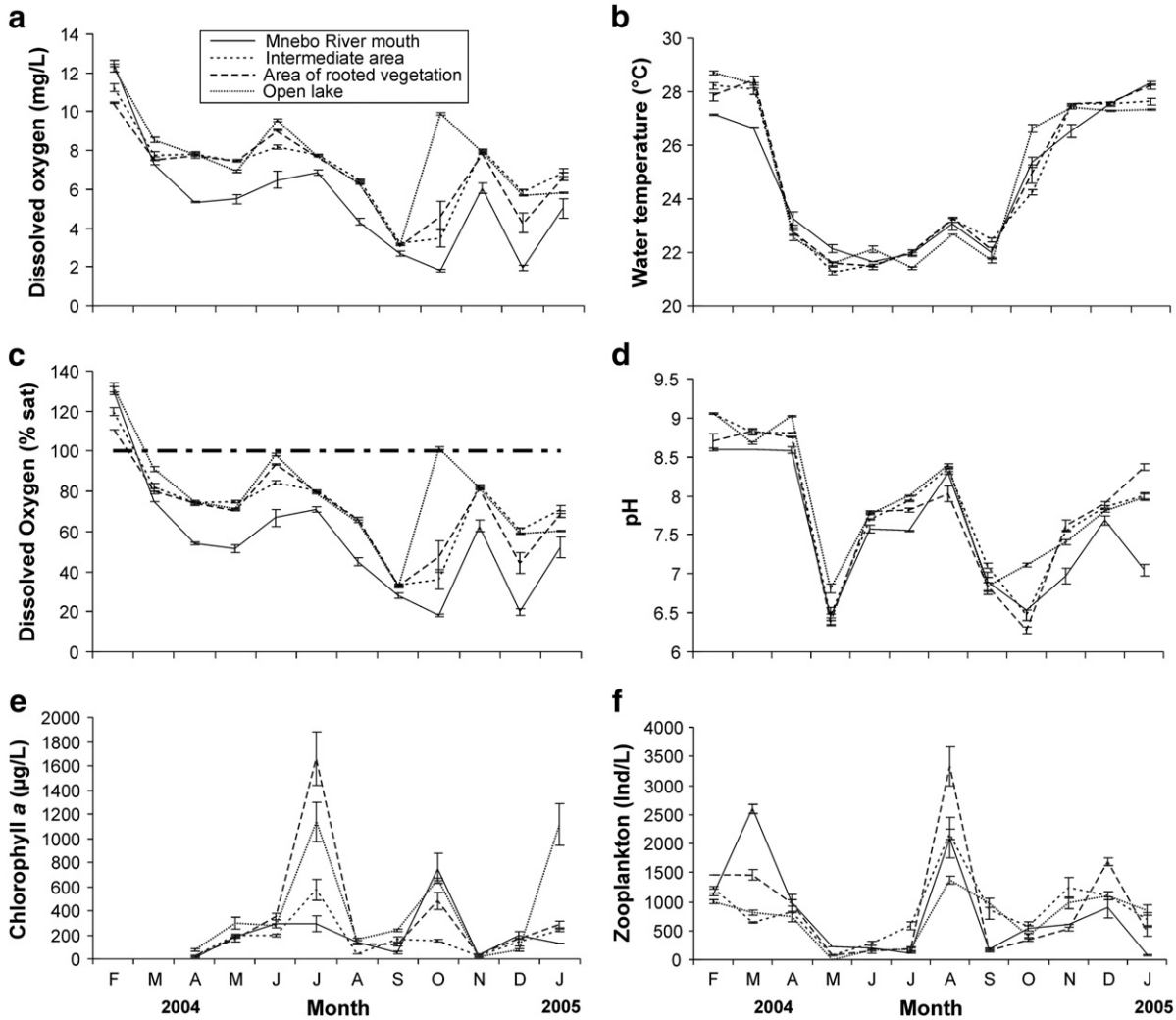

Fig. 4. Seasonal variation of (a) dissolved oxygen, (b) water temperature, (c) dissolved oxygen saturation, (d) pH, (e) chlorophyll a, and (f) zooplankton. Mean ± SE.

As a closed basin lake, ionic concentrations as measured by electrical conductivity (EC) are relatively high and variable over the season. The seasonal rains increased the river flow and caused a dilution pulse in the lake especially in the vicinity of the river mouth where electrical conductivity dropped rapidly with rising river flow during the rainy season. The maximum conductivity of 2677 µS/cm was reached in October at the time of minimum relative water level, and minimum values were recorded in December just before the lake achieved its observed maximum level in January. A sharp decrease in conductivity was observed with the onset of highest rainfall in December, when the river mouth reached the minimal values of about 383 µS/cm. In December, a gradient of conductivity was seen from the river mouth, through the swamp and inshore zones to the open lake, with mean conductivity for the different zones recorded as 383, 1226, 1507 and 2069 µS/cm respectively (Fig. 3e). After the end of the rainy season, the gradual rise in conductivity from April to October coincided with the period of no rainfall; and the rise was consistent with the decline in relative water levels due to evaporation. During this low water period, conductivity was relatively uniform among the sampling sites indicating strong horizontal circulation among the sites.

Dissolved oxygen, dissolved oxygen saturation, and pH

The highest dissolved oxygen (DO) level at all sites was recorded in February (Fig. 4a); however, the river mouth regularly registered lower concentrations compared to other areas while following a similar temporal trend over the annual cycle. From the February high concentrations, DO then decreased at all sampling sites, but remained relatively high before further declining during the cooler, dry months of May to September. Low and similar values of about 3.2 mg/L were measured in September in all zones. While the DO away from the river mouth increased in October to much higher concentrations in intermediate, area of rooted vegetation and open lake were much higher at 4.6, 3.5 and 9.9 mg/L respectively, the DO continued decreasing in the river mouth to a minimum of 1.8 mg/L, well below atmospheric saturation. In November the DO in intermediate areas and open lake was similar (8.0 mg/L) and close to atmospheric saturation, while in the river mouth and area of rooted vegetation the concentrations were undersaturated at 2 mg/L and 4 mg/L respectively. A second decrease in all stations was observed in December, again with the lowest values again recorded in the river mouth while electrical conductivity was also at its minimum.

Dissolved oxygen saturation followed a similar pattern to dissolved oxygen concentration with high values in February and minimum values in September in all sampling sites with the exception of the river mouth where it continued to decline until October. With the exception of three distinct peaks (February, June and October when the open lake was at or above atmospheric saturation, Lake Chilwa waters remain below 100% saturation with the atmosphere (Fig. 4c). Low values, below 60%, were measured in September in all zones when relative lake level was approaching the annual minimum.

While the dissolved oxygen saturation in the intermediate areas, area of rooted vegetation and open waters increased in September, the dissolved oxygen saturation continued decreasing in the Mnembo River mouth and area of rooted vegetation falling well below 60% saturation into October when the saturation levels in intermediate and open lake were above 60% saturation.

The pH showed a defined seasonal pattern (Fig. 4d), with maxima exceeding 8.5 between February and April. Following the end of the rains, and as the relative lake level fell and suspended solids rose, pH declined to minimum values (<7) in October. The onset of rainfall then coincided with a pH rise from <7 to ~8 by January as suspended sediment concentrations fell. pH was lowest in the river mouth on eight of 12 samplings. The major declines of pH in river were observed in March and October; with the lowest pH of 6.3 recorded in the swamp zone in October. In general there was coherence between the percent of atmospheric saturation and pH, with both parameters showing three broadly contemporaneous maxima (Fig. 4c and d).

Total suspended solids, windspeed, chlorophyll a, and zooplankton

All sampling sites showed a similar pattern in total suspended solids, with maxima at the end of the dry season. In contrast, minimum TSS concentrations were recorded during the rainy season (November to April) when TSS was below 0.2 mg/L (Fig. 3f). At this time, the river mouth recorded lower average TSS than the lake. TSS increased gradually from June, reaching high values above 1 mg/L in September in all sampling sites, coincident with the shallowest relative lake level and a period of high wind speeds (Fig. 3d). TSS then decreased abruptly from October (Fig. 3f), coincident with the onset of the rains with all sampling sites returning to low concentrations similar to the preceding rainy season. Highest concentrations of TSS coincided with the highest wind speeds in September–October during the low stand (Fig. 3c and a). However, an earlier peak in wind speed in March–April seemed not to have a strong effect on TSS when the lake was at higher level. TSS started to rise as relative lake levels fell even though wind speeds were not exceptional suggesting that sediment resuspension was requiring less mechanical energy input as lake level dropped. Little variation of wind speed was observed in the dry season of June, July and August. Lowest wind speeds were in the dry season and during the rainy season with higher winds associated with the transitional periods in these two seasons which are associated with the passage of the Intertropical Convergence Zone over Lake Chilwa.

L. Chilwa is rich in phytoplankton with chlorophyll concentrations in the open lake always above 15 μg/L. The abundance of phytoplankton, as represented by chlorophyll-*a* concentration, varied considerably by location and did not follow a generalized and simple seasonal pattern (Fig. 4e). There were three clear peaks in chlorophyll during the study period with increases in July, October and January, followed in each case by a period of relatively low concentration. The area of rooted vegetation showed a similar pattern of variability to the open lake and intermediate areas, except that there was no clear January peak when the river plume may have been the dominant water mass in these vegetated area.

As with chlorophyll-*a*, the abundance of zooplankton fluctuated as a series of peaks throughout the year and did not follow a simple seasonal pattern (Fig. 4e and f). The first peak occurred in March with high abundance in the river mouth at 2600 individuals/L while lower abundance less than 850 ind./L was recorded in intermediate areas and open waters. The second peak occurred in August, following the main chlorophyll-*a* peak in July, with the highest abundance recorded in swamps (3333 ind./L). At the same time in the river mouth, intermediate areas and open waters ranged between 1373 and 2100 ind./L. The third peak occurred in December, again with highest abundance in the areas of rooted vegetation (1693 ind./L). The abundance in intermediate areas and the open lake was similar (1107 ind./L), but was higher than the river mouth (900 ind./L). Soon after the peak in December, the abundance decreased to less than 1000 ind./L in all areas.

Discussion and conclusion

A striking feature of the Mnembo River mouth region of Lake Chilwa is the extent to which the open lake, the areas of rooted vegetation and the river mouth contain very similar water for much of the year. All zones exhibited a temporally coherent and similar seasonal variation in their physico-chemical characteristics. In contrast, variability in phytoplankton and zooplankton was more complex and did not appear to be related in a simple way to any single meteorological or physico-chemical factor. During the dry season (May–September), relative water levels fell; all sampling sites had cooler water temperatures with declining pH and dissolved oxygen saturation which reached very low values, below 60%, in September at all sites indicating low productivity relative to respiration. Similarly, low values below 60% dissolved oxygen saturation, were recorded in December in Mnembo River mouth and the area of rooted vegetation after the heavy rain began and relative water levels rose rapidly. In general, the river mouth was well below atmospheric equilibrium and the lowest of the sampled sites indicating that respiration predominated over photosynthesis and atmospheric exchange. During the dry season, the electrical conductivity increased by evaporative concentration and the resulting lowering water level enabled easier resuspension of bottom sediments causing increases in suspended solids. The onset of rainfall in October caused rising water level and a reversal of the prior trends in suspended solids and electrical conductivity. The rains and rising water levels after October were also associated with increasing pH and improved oxygenation of the water column especially in the open lake which may also be indicative of improved conditions supporting higher rates of primary production. The observed peak in chlorophyll in October is consistent with this interpretation.

The general pattern we observed in the hydrologic budget agrees with that found by Morgan and Kalk (1970). They observed that EC in Lake Chilwa follows a dramatic seasonal pattern, and the peaks and troughs observed during the dry and wet season are attributed to alternatively to evaporative concentration and rainfall. The dilution effect of the rains on EC in Lake Chilwa (Morgan and Kalk, 1970) was confirmed in the present study which demonstrated that the river itself also experienced seasonal changes in EC. However, the higher EC level recorded in February compared to January one year later is believed to be a consequence of higher EC level in2003, the previous year. We used the same instrument throughout the study and we believe that there were no instrument errors.

The strong seasonality of rainfall and temperature at Lake Chilwa is thus a powerful driver of lake conditions. Insofar as conductivity at the river mouth sites is an indicator of flow of freshwater from the catchment, it was not until December that a substantial pulse of catchment-derived began to freshen the lake. Thus it appears that early rainfall from October to November was recharging the catchment and the lake margin wetland reservoirs rather than the river mouth and lake, and the small decline in lake conductivity and increase in lake level in October-November is more likely due to the effects of direct rainfall to the lake and marginal swamps and flushing of water from lake marginal vegetation (Howard-Williams, 1972) than fresh water directly from the Mnembo River. The low EC values in the river mouth in December, reached after the relative lake level started to rise are an indication of the low dissolved ion concentrations in the flooding river and this freshening was accompanied by low oxygen concentrations indicating that the river was also carrying a high organic load from the catchment. Low oxygen levels in river mouth may also be attributed to the influx of bottom water flushed from the swamp carrying a plume of swamp detritus, nutrients and mineral ions into the lake (Boar, 2006) creating hypoxic conditions common in dense swampy tropical freshwaters (Chapman et al., 2004). The quantity of materials

exchanged between the swamp and the lake depends on the size of the swamp. Rivers passing through large swamps will transport more swamp detritus compared to one that runs in a small catchment (Boar, 2006) suggesting that Mnembo River passes through a large swamp area before reaching the lake.

It was expected that the particulate materials flushed into the sampling sites during the rainy season could immediately increase TSS concentrations from the river mouths outwards. This was not the case. The generation of high TSS concentrations occurred during the dry season when the combination of low lake level and strong persistent winds (Fig. 3c, d, and f) facilitates sediment resuspension. The sharp drop in TSS in October is suggested to be attributed to reduced wind stress and increase in lake that accompanied the onset of the rains.

Physical, climate-related processes play a major role in the dynamics of L. Chilwa. In this study we have confirmed their importance in driving aspects of seasonal variability that in terms of EC is comparable to the seasonal cycle described three decades ago. Set against strong seasonality in many physical and chemical variables, phytoplankton and zooplankton show what appears to be a more dynamic variability, with short-lived peaks and troughs throughout the year likely dependent on the availability of light to drive photosynthesis, but with consistent spatial effects across habitats. What controls the cycles of phytoplankton and zooplankton? There is no evidence that they are simply related to temperature or rainfall cycles, yet the correspondence among the habitat types examined here implies that they are under control of number of interacting and physically driven variables. The alternating peaks of abundance of phytoplankton and zooplankton are suggestive of an oscillation driven by periods of increased algal growth induced a strong grazing response by the zooplankton which then reduced phytoplankton. A peak of phytoplankton was followed by zooplankton peak in the following month or two (Fig. 4e and f). The first and second phytoplankton peaks were recorded in July and October respectively while for zooplankton, the peaks were observed in March, August and December. We hypothesize that if chlorophyll sampling had been initiated at the beginning of our sampling period we would also have observed a chlorophyll maximum in January or February in response to nutrient inputs and low TSS during the rainy season. Such a peak would have fed the zooplankton maximum that we observed in March. The generally high and remarkably high peak concentrations of chlorophyll suggests high primary production much of the year which would be consistent with historic high fish productivity of Lake Chilwa (Furse et al., 1979) of about 344 kg/ha/year (Turner, 1996).

The fish productivity of shallow L. Chilwa is high. The foundation of this high productivity is not only high rates of autochthonous oxygenic production of the abundant phytoplankton but also may benefit from oxygen consuming heterotrophic production driven by allochthonous organic inputs arising from the fringing wetlands and catchment (Howard-Williams and Lenton, 1975). The oxygen balance in open waters reflects these two processes. Over most of the year, the lake was below oxygen saturation despite being a large, shallow windy lake, factors that should favor rapid equilibration of the shallow water column with the atmosphere. In addition, the lake has high chlorophyll levels, generally exceeding 100 μg/L and as high as 1000 μg/L, indicative of high autochthonous primary productivity and oxygen production. Evidently heterotrophic bacterial production is so high that despite expected high rates of oxygen production, full equilibration with the atmosphere is not generally realized. In fact the periods of near saturation or super-saturation, February, June and October were followed or were contemporaneous with evidence of increased biological production for example zooplankton peaks in 2004 in March, August and December which in turned followed chlorophyll peaks in July, October–November. There was also a chlorophyll peak in January 2005 and if this were annually recurring a similar peak may have accompanied the maximum in oxygen saturation February of 2004 (when chlorophyll was not measured). These peaks in oxygen saturation and chlorophyll were also coincident or near-coincident with periods of elevated pH which would result from high levels of carbon uptake by the inferred high levels of photosynthesis. Despite generally high chlorophyll levels that are indicative of high rates of primary productivity, the input of organic materials from the catchment and especially the extensive fringing wetlands also supports high rates of bacterial oxygen consumption. Thus bacterial production may contribute substantially to secondary and fish productivity if there is a microbial food web capable of transferring this bacterial productivity to the upper food web.

The three observed chlorophyll and zooplankton maxima and periods of oxygen saturation and super-saturation in 2004–2005 clearly show that the lake can achieve high levels of primary productivity for periods of time. An important question is whether these maxima are truly periodic being the product of predictable aspects of the physical environment or are they aperiodic and inherently chaotic depending on circumstantial interactions of numerous factors. We tentatively suggest that they may be periodic but future studies are required to fully answer these questions. If we follow a hydrological year and starting with the rainy season, the first maximum in primary productivity occurs during January–March coincident with high rainfall and high relative lake levels. The river brings a nutrient load into a deeper transparent lake (low TSS), favoring primary productivity resulting in oxygen super-saturation, high pH (owing to high uptake of CO_2) increased chlorophyll (observed in 2005; no measurement in 2004). This primary productivity peak is succeeded by rapid growth in the zooplankton to their peak abundance, which may overgraze the phytoplankton leading to a clear water phase of lower chlorophyll, lower oxygen and lower TSS in April.

With cessation of the rains in May and reduced river flow, nutrient inputs decline and lake levels drop. This declining water level initially results in a second pulse of primary and secondary production as benthic stirring of lake sediments returns nutrient to the water column originally deposited during the rainy season. This second pulse occurs while the water level is dropping but suspended sediment concentrations are still low allowing light penetration to be adequate for phytoplankton to take up the nutrients returned from the bottom and results in a second maximum in O_2 saturation, pH, chlorophyll and eventually zooplankton. This second productive phase is terminated again by zooplankton grazing but also by falling primary productivity rates as the lake becomes so shallow that sediment resuspension increases TSS to its maximum in September and lowers light penetration so that phytoplankton photosynthesis becomes light limited resulting in the lowest O_2 saturation values, lowest pH and low chlorophyll levels in September.

Following the period of minimum productivity in September, the declining trends were reversed in October when a decline in TSS concentration was initiated. At this time, the lake underwent a rapid warming (Fig. 4b) which may have resulted in not only diurnal but also more persistent diel stratification of the water column which may persist when the lake is warming rapidly and which would reduce resuspension of sediments from the bottom relieving light-limitation and allowing higher rates of photosynthesis. All the indicators of primary productivity, O_2 saturation, pH and chlorophyll move upward in October and higher zooplankton abundances follow in November and December. In December there is also a rapid rise in lake level restarting the hydrological year and the productivity succession we hypothesize.

The three proposed seasonal pulses of plankton production are not explained by anyone factor but rather the interaction of the initial pulse of riverine nutrient inputs into clear water at high lake level which begins the annual cycle, a second pulse as lake levels decline that is nourished by benthic regeneration of nutrients and a third pulse that is initiated by a rapid warming of the water column and inferred

thermal stratification. It is also possible that intense zooplankton grazing imposes a top down limitation on phytoplankton abundance which, in turn, may be controlled by the seasonal abundance cycle of zooplanktivorous fish (Kalk, 1979). The reality of this successional sequence which results in the high primary productivity of the lake needs to be tested with independent and more detailed measurements over the annual cycle. Direct measurements of primary production and water column respiration would also clarify the role of heterotrophic production in the lakes annual cycle.

Lake Chilwa is a significant lake ecologically as well as economically. Its support of a wide range of wetland species is attested to by its RAMSAR status (Environmental Affairs Department, 2000) and its contribution to national Malawian fishery production is substantial. This high productivity seems to be in part supported by the connectivity between the lake, its marginal wetlands and its inflow river and their deltas. Managing this lake needs to take into account its sensitivity and dependence on these interconnections to maintain fish productivity. Clearly, this shallow and saline lake is unique, and management plans must take its highly variable water budget into consideration. Human land use in the surrounding catchment (Jamu et al., 2003) will affect L. Chilwa differently than other freshwater bodies especially if agricultural pressures lead to loss of the productive fringing wetlands or increase the sediment load to a system where primary productivity is likely already light limited much of the year. If this situation is not counteracted by management policies, there will indeed be decline in the productive potential of the fishery which supports 335 villages in Malawi and contributes about 20% of all fish caught in Malawi. In addition, we suggest that for sustainable exploitation and conservation of L. Chilwa resources, the management strategy should also take into consideration existing livelihood strategies developed by L. Chilwa people over decades in relation to their dynamic response to historic climate change which also had direct effect on relative water levels, the physico-chemical parameters, biological productivity, fish species composition and annual fish catches.

The fishing families of Lake Chilwa have long exploited the lake and its catchment in a sustainable manner by being able to switch their activities to take advantage of the nutritional benefits of a productive fishery when it was available but returning to farming when the lake was dry or unproductive. The risk now is that the productive fishery may slowly be lost without careful land management and wetland protection. The specter of climate change and increasing aridity could greatly accelerate the pace of change for the naturally astatic lake ecosystem. An integrated approach to catchment management that recognizes the long established patterns of livelihoods dependent on both lake and land will be necessary if the people are to continue to thrive or even survive.

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